

ElectroHub Retail Sales & Profitability Analytics Dashboard

Project Overview

- Company
 - ElectroHub
 - Multi category retail company
- Product Categories
 - Electronics
 - Clothing
 - Footwear
 - Home Appliances
 - Accessories
 - Kitchenware
 - Bags
 - Personal Care
- Goal
 - Centralized reporting
 - Sales monitoring
 - Profitability tracking
 - Discount analysis
 - Regional sales analysis
 - Interactive BI dashboard

Business Requirements

- Product Performance
 - Top 5 Products by Sales
 - Bottom 5 Products by Sales
 - Top 5 Products by Profit
 - Bottom 5 Products by Profit
 - Top 5 Products by Quantity Sold
 - Bottom 5 Products by Quantity Sold
- Sales Trends
 - Daily Sales
 - Monthly Sales
 - Quarterly Sales
 - Yearly Sales
- Profitability Analysis
 - Sales vs Profit Relationship
- Comparative Analysis
 - Compare two date ranges
 - Dynamic date selection
- Discount Analysis
 - Average Discount by Promotion
- Operational KPIs
 - Total Orders
 - Quantity Sold
- Geographic Analysis
 - Net Sales by City
- Detailed Reporting
 - Product Filter
 - Customer Filter
 - Date Filter
 - Promotion Filter

Technology Stack

- Database
 - Microsoft SQL Server
- Data Processing
 - Power Query
- Data Modeling
 - Star Schema
- BI Tool
 - Power BI Desktop
- Calculations
 - DAX

Dataset Overview

- Customer Table
 - Fields
 - Customer ID
 - Customer Name
 - City
 - State
 - Pincode
 - Email ID
 - Phone Number
 - Purpose
 - Customer analysis
 - City analysis
- Product Table
 - Fields
 - Product ID
 - Product Name
 - Product Line
 - Price
 - Purpose
 - Product performance
 - Category analysis
- Promotion Table
 - Fields
 - Promotion ID
 - Promotion Name
 - Ad Type
 - Coupon Code
 - Price Reduction Type
 - Purpose
 - Discount analysis
 - Promotion analysis
- Sales Transactions Table
 - Fields
 - Date
 - Customer ID
 - Promotion ID
 - Product ID
 - Units Sold
 - Price Per Unit
 - Total Sales
 - Discount Percentage
 - Discount Value
 - Net Sales
 - Purpose
 - Fact table
 - KPI calculations

Environment Setup

- ElectroHub_Test
 - Records
 - ~300 transactions
 - Purpose
 - KPI validation
 - Relationship testing
 - DAX testing
 - Dashboard testing
- ElectroHub_Prod
 - Records
 - 3000+ transactions
 - Purpose
 - Production reporting
 - Final dashboard deployment

Data Cleaning & Transformation

- Data Type Validation
 - Verify all column types
 - Numeric fields
 - Date fields
 - Text fields
- Discount Transformation
 - Source
 - Price Reduction Type
 - Examples
 - 30% Off
 - Buy 1 Get 1 Free
 - Action
 - Create Discount Percentage column
- Merge Operations
 - Merge
 - Product Table
 - Sales Transactions
 - Key
 - Product ID
 - Result
 - Populate Price Per Unit
- Missing Value Handling
 - Columns
 - Discount Percentage
 - Action
 - Replace Null with 0
- Calculated Columns
 - Total Sales
 - Units Sold x Price Per Unit
 - Discount Value
 - Total Sales x Discount %
 - Net Sales
 - Total Sales - Discount Value
 - Profit
 - 10% of Net Sales

Data Modeling

- Schema Design
 - Star Schema
- Fact Table
 - Sales Transactions
- Dimension Tables
 - Customer
 - Product
 - Promotion
- Relationships
 - Customer → Sales
 - Customer ID
 - One-to-Many
 - Active
 - Product → Sales
 - Product ID
 - One-to-Many
 - Active
 - Promotion → Sales
 - Promotion ID
 - One-to-Many
 - Active
- Benefits
 - Simplified filtering
 - Better performance
 - Scalable model

DAX Measures

- Profit Measure
 - Formula
 - Profit = 10% of Net Sales
- Comparative Period Analysis
 - Components
 - Date Table 1
 - Date Table 2
 - Concept
 - Inactive relationship
 - Function Used
 - USERELATIONSHIP()
 - Purpose
 - Compare two date periods

Dashboard Development

- KPI Cards
 - Total Sales
 - Net Sales
 - Total Profit
 - Total Orders
 - Quantity Sold
 - Average Discount
- Product Performance Visuals
 - Top 5 Products
 - Bottom 5 Products
 - Metrics
 - Sales
 - Profit
 - Quantity
- Sales Trend Visuals
 - Daily Trend
 - Monthly Trend
 - Quarterly Trend
 - Yearly Trend
- Profitability Analysis
 - Visual
 - Scatter Plot
 - Analysis
 - Sales vs Profit
- Discount Analysis
 - Visual
 - Bar Chart
 - Metric
 - Average Discount by Promotion
- Geographic Analysis
 - Visual
 - Map
 - Metric
 - Net Sales by City
- Detailed Reporting
 - Filters
 - Product
 - Customer
 - Promotion
 - Date

Validation & Deployment

- Testing Activities
 - KPI Verification
 - Relationship Validation
 - Filter Testing
 - DAX Validation
 - Visual Testing
- Production Migration
 - From
 - ElectroHub_Test
 - To
 - ElectroHub_Prod
 - Records
 - 300 → 3000+
 - Outcome
 - Final dashboard deployment